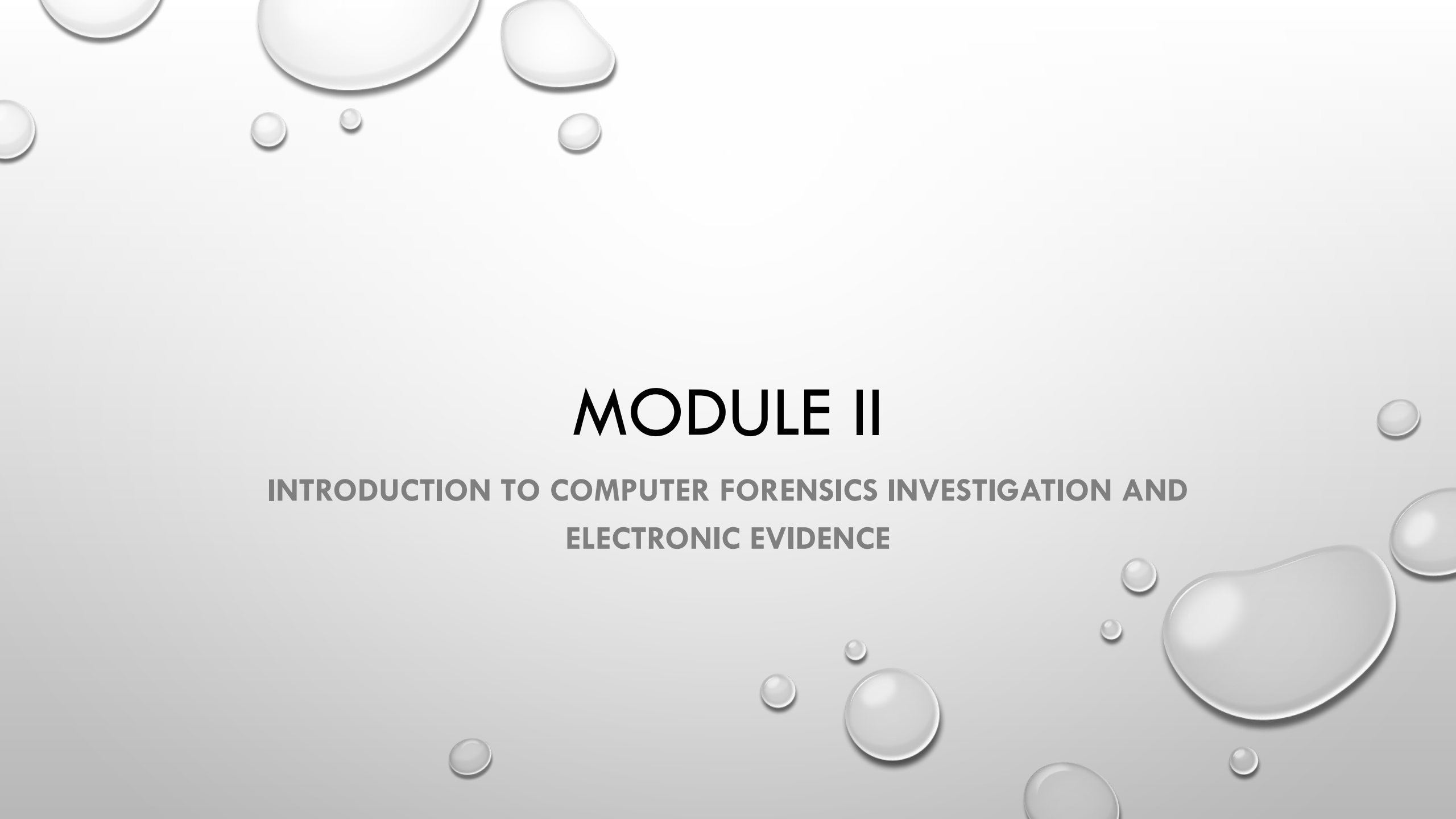


The background of the slide is a light gray gradient. It is decorated with numerous realistic water droplets of various sizes. Some droplets are large and prominent, while others are small and scattered. They are primarily located in the top-left, bottom-right, and bottom-center areas, leaving the central text area clear.

MODULE II

**INTRODUCTION TO COMPUTER FORENSICS INVESTIGATION AND
ELECTRONIC EVIDENCE**

IMAGING AND CLONING

- **COPYING**

- ☐ COPYING MEANS **DUPLICATING DATA OR FILES FROM ONE LOCATION TO ANOTHER.**
- ☐ WHEN ONE DOES COPYING, IT IMPLIES THAT ONE IS **USING STANDARD COPY COMMANDS.**
- ☐ HOWEVER, WHEN DEALING WITH EVIDENCE, IT MIGHT BE HARD TO USE COPY ONLY.
- ☐ IT IS BECAUSE THE STANDARD COPY CAN ALTER THE METADATA AND CHANGE THE HIDDEN OR DELETED DATA .
- **THE CHARACTERISTICS OF COPYING INCLUDE:**
 - **SPEED:** COPYING IS SIMPLER AND FASTER, COMPARED TO CLONING OR IMAGING.
 - **RISK:** THE RISK INVOLVED IN COPYING IS THAT THE METADATA MIGHT BE ALTERED AND ALL THE DATA MIGHT BE CAPTURED.

- **CLONING**
- ❑ IT IS THE PROCESS WHERE THE **TRANSFER OF THE ENTIRE CONTENTS OF A HARD DRIVE OR A STORAGE DEVICE IS DONE ON ANOTHER STORAGE DEVICE** . THIS PROCESS IS KNOWN AS CLONING.
- **CHARACTERISTICS OF CLONING:**
- **BIT-FOR-BIT REPLICATION:** CLONING KEEPS THE EXACT CONTENT AND THE WHOLE STRUCTURE OF THE ORIGINAL DEVICE.
- **USE CASES:** CLONING IS USED WHEN IT IS NEEDED TO KEEP THE ORIGINAL DEVICE INTACT FOR FURTHER EXAMINATION OR A LEGAL AFFAIR.
- **TIME CONSUMING:** CLONING IS USUALLY LONGER IN COMPARISON TO SIMPLE COPYING SINCE IT INVOLVES THE WHOLE DETAILED REPLICATION. THOUGH IT **DEPENDS ON VARIOUS FACTORS LIKE THE SIZE OF THE STORAGE DEVICE, THE SPEED OF THE DEVICES INVOLVED, AND THE METHOD OF CLONING.**

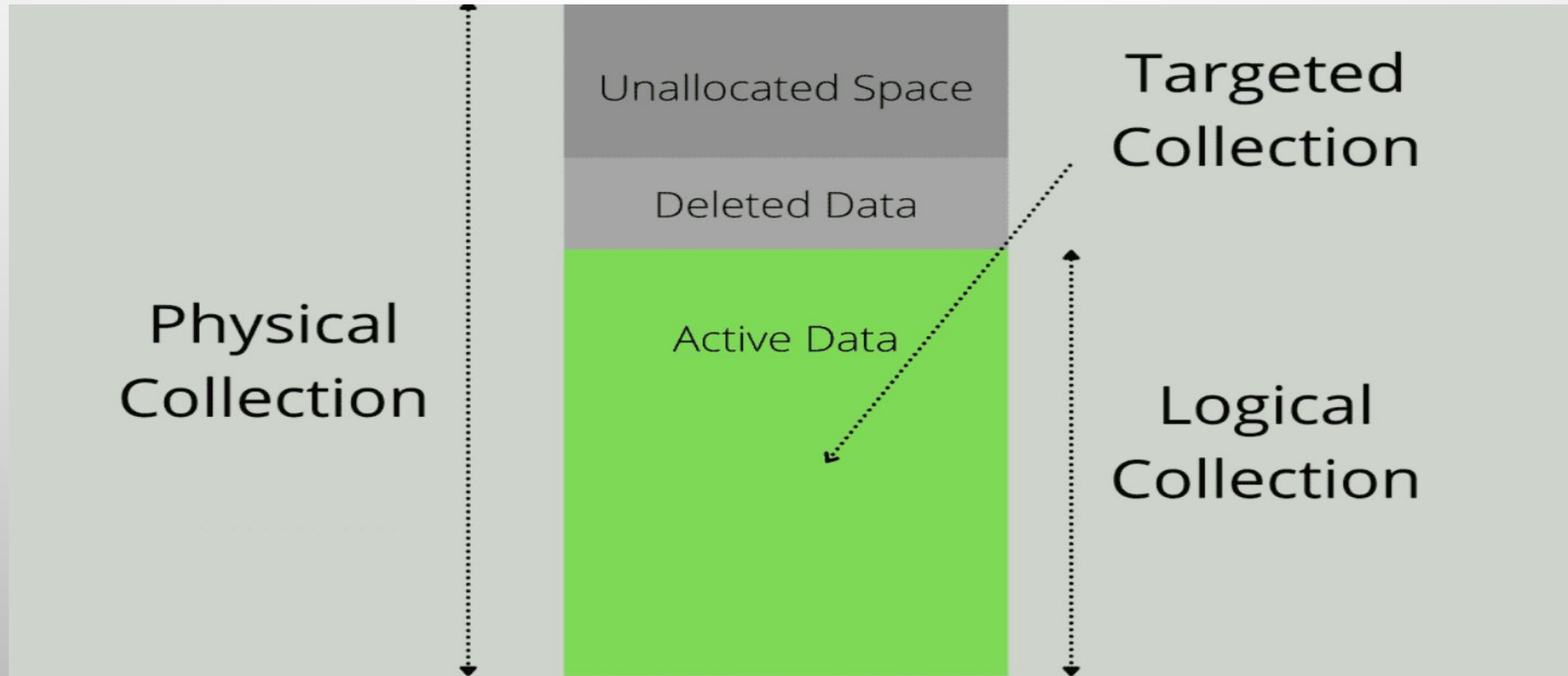
- **IMAGING:**

- ❑ IT IS THE **PROCESS OF CREATING A FORENSIC IMAGE OF A STORAGE DEVICE.**
- ❑ A FORENSIC IMAGE IS A **REPLICA COPY OF EVERY BIT OF DATA THAT WAS ON THE SOURCE DEVICE, THIS INCLUDING THE ALLOCATED, UNALLOCATED, AND THE AVAILABLE SLACK SPACE .**
- ❑ THE IMAGE IS THEN USED FOR ANALYSIS AND INVESTIGATION, AND THE ORIGINAL EVIDENCE IS LEFT UNTOUCHED.

- **CHARACTERISTICS OF IMAGING:**

- **INTEGRITY:** IMAGING ENSURES THE INTEGRITY AND AUTHENTICITY OF THE EVIDENCE PRODUCED.
- **FLEXIBILITY:** FORENSIC IMAGE REPLICAS CAN BE MOUNTED AS A VIRTUAL DRIVE TO CREATE IMAGE-SPECIFIC MODE FOR ANALYSIS OF DATA WITHOUT AFFECTING THE ORIGINAL EVIDENCE .
- **METADATA:** IMAGING CAPTURES METADATA ASSOCIATED WITH THE DATA, THUS PROMOTING FORENSIC ANALYSIS.

TYPES OF IMAGING



PHYSICAL IMAGING

- ❑ A PHYSICAL IMAGING IS A BIT-BY-BIT COPY OF THE DEVICE, I.E., AN EXACT COPY.
- ❑ CONDUCTING PHYSICAL IMAGING IS THE MOST THOROUGH AND ACQUIRES THE GREATEST AMOUNT OF DATA.
- ❑ IT IS USED TO ACQUIRE THE ENTIRE PHYSICAL VOLUME OF A DRIVE.
- ❑ PHYSICAL FORENSIC IMAGES CAPTURE DELETED SPACE, FILE FRAGMENTS AND PROVIDES ACCESS TO DELETED AND ENCRYPTED DATA.
- ❑ THE PROS: FULL ACCESS TO DEVICE ARTIFACTS (E.G. EVENT LOGS, FILES, AND TIMESTAMPS).
- ❑ THE CONS: MORE DATA = MORE MONEY.

LOGICAL IMAGING

- ❑ A LOGICAL IMAGE OF A DEVICE OR HARD DRIVE CAPTURES ALL FILES VISIBLE TO THE USER AND TYPICALLY **DOES NOT RECOVER DELETED ITEMS, DATA IN DELETED AREAS OF THE DEVICE, NOR DOES IT COLLECT FILE FRAGMENTS.**
- ❑ BASICALLY, A LOGICAL DEVICE COLLECTION **ISOLATES ONLY “ACTIVE” FILES ON A DEVICE.**
- ❑ PROS OF LOGICAL IMAGING: **LESS DATA AND LESS EXPENSIVE.**
- ❑ CONS: **DOES NOT PROVIDE MUCH INSIGHT INTO DELETED FILES.**

TARGETED IMAGING

- ❑ A FORENSIC COLLECTION OF SPECIFIC FILES OR FOLDERS RELEVANT TO A LEGAL MATTER.
- ❑ THIS METHOD IS THE LEAST EXPENSIVE BECAUSE IT COLLECTS THE LEAST AMOUNT OF DATA AND IS BEST SUITED FOR CIVIL MATTERS WITH COST SENSITIVITY CONSIDERATIONS.
- ❑ PROS OF TARGETED FORENSIC COLLECTIONS: SPEEDS UP E-DISCOVERY PROCESS BECAUSE OF LESS PROCESSING TIME.
- ❑ CONS: MAY NEED TO GO BACK TO COLLECTION SOURCE IF ADDITIONAL DATA IS NEEDED THAT WAS NOT IN ORIGINAL TARGETED COLLECTION.

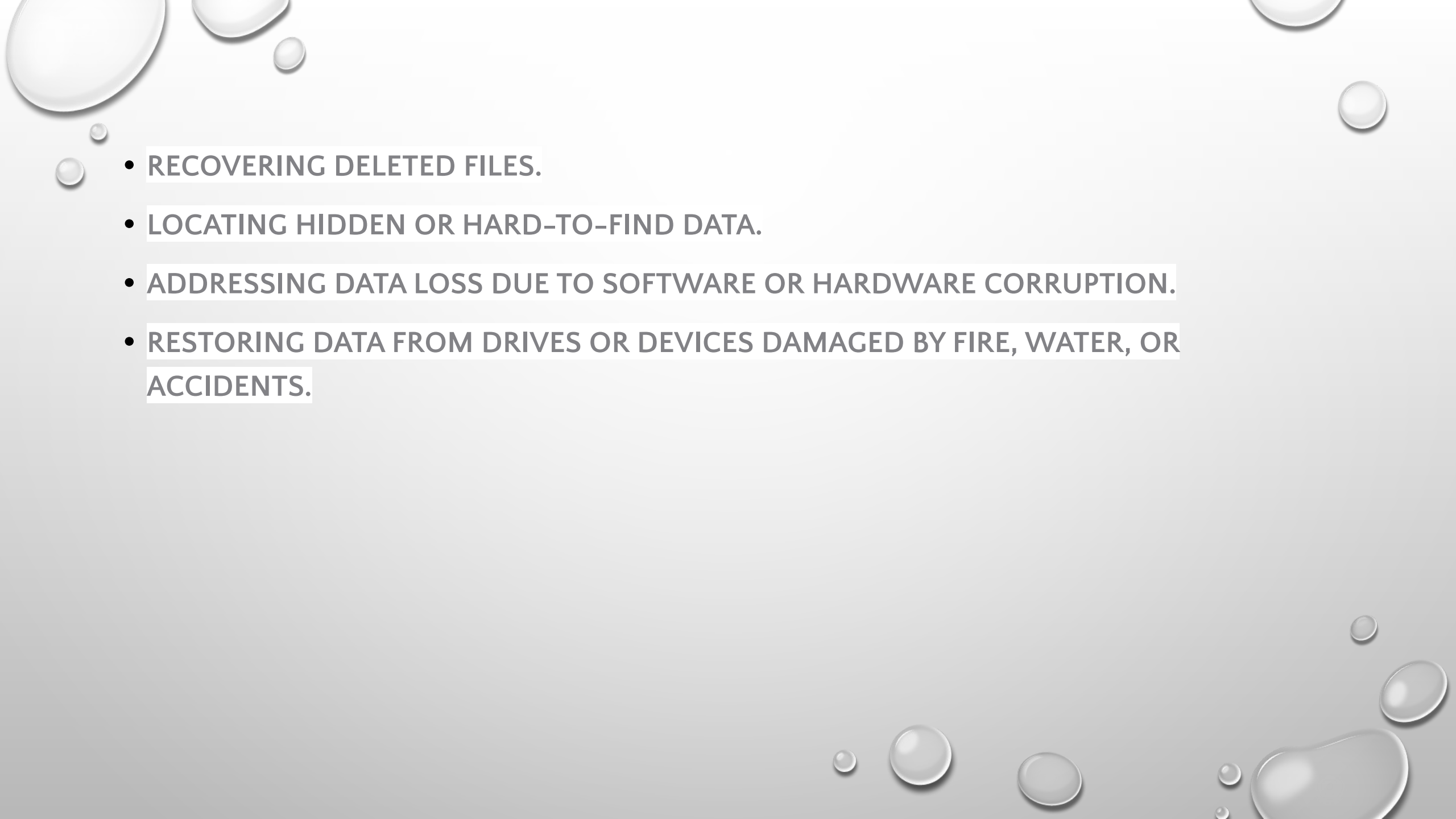
DATA RECOVERY

- DATA RECOVERY IS A CRUCIAL ASPECT OF DIGITAL FORENSICS, THE PROCESS OF COLLECTING AND ANALYZING DIGITAL EVIDENCE.
- **IN TODAY'S DIGITAL WORLD, WHERE SO MUCH OF OUR LIVES ARE STORED ON ELECTRONIC DEVICES, THE ABILITY TO RECOVER DELETED OR HIDDEN DATA IS ESSENTIAL FOR INVESTIGATIONS OF ALL KINDS, FROM CYBERCRIME TO CORPORATE FRAUD.**
- **FORENSIC DATA RECOVERY IS A SPECIALIZED FIELD THAT INVOLVES RETRIEVING HIDDEN OR LOST DATA FROM DIGITAL DEVICES.**
- THIS DATA CAN INCLUDE **DELETED FILES, EMAILS, INTERNET BROWSING HISTORY, AND MORE.**
- FORENSIC DATA RECOVERY IS DIFFERENT FROM TRADITIONAL DATA RECOVERY IN THAT IT MUST BE DONE IN A **WAY THAT PRESERVES THE INTEGRITY OF THE EVIDENCE.**
- **THIS MEANS THAT THE DATA MUST BE RECOVERED WITHOUT ALTERING OR MODIFYING IT IN ANY WAY.**

WHY IS DATA RECOVERY IMPORTANT IN DIGITAL FORENSICS?

DATA RECOVERY CAN BE USED FOR A VARIETY OF PURPOSES IN DIGITAL FORENSICS, INCLUDING:

- **COLLECTING EVIDENCE IN CRIMINAL INVESTIGATIONS:** DELETED FILES OR HIDDEN DATA CAN PROVIDE VALUABLE EVIDENCE IN CRIMINAL CASES, SUCH AS CHILD PORNOGRAPHY CASES OR MURDER INVESTIGATIONS.
- **RECOVERING LOST DATA IN CIVIL LITIGATION:** DATA RECOVERY CAN BE USED TO RECOVER LOST DATA IN CIVIL CASES, SUCH AS BUSINESS RECORDS OR INTELLECTUAL PROPERTY.
- **CONDUCTING CORPORATE INVESTIGATIONS:** DATA RECOVERY CAN BE USED TO INVESTIGATE EMPLOYEE MISCONDUCT OR CORPORATE FRAUD.

- 
- RECOVERING DELETED FILES.
 - LOCATING HIDDEN OR HARD-TO-FIND DATA.
 - ADDRESSING DATA LOSS DUE TO SOFTWARE OR HARDWARE CORRUPTION.
 - RESTORING DATA FROM DRIVES OR DEVICES DAMAGED BY FIRE, WATER, OR ACCIDENTS.

BASICS OF DATA DELETION

- THE PRINCIPLE OF FILE RECOVERY OF DELETED FILES IS BASED ON THE FACT THAT WINDOWS DOES NOT WIPE THE CONTENTS OF THE FILE WHEN IT'S BEING DELETED.
- INSTEAD, A FILE SYSTEM RECORD STORING THE EXACT LOCATION OF THE DELETED FILE ON THE DISK IS BEING MARKED AS “DELETED” AND THE DISK SPACE PREVIOUSLY OCCUPIED BY THE DELETED FILE IS THEN LABELLED AS AVAILABLE – BUT NOT OVERWRITTEN WITH ZEROES OR OTHER DATA.



- A ***DELETED FILE*** IS ANY FILE THAT HAS BEEN LOGICALLY ERASED FROM THE FILE SYSTEM BUT MAY STILL REMAIN PHYSICALLY ON STORAGE MEDIA.

- HOW A FILE IS DELETED CAN VARY:

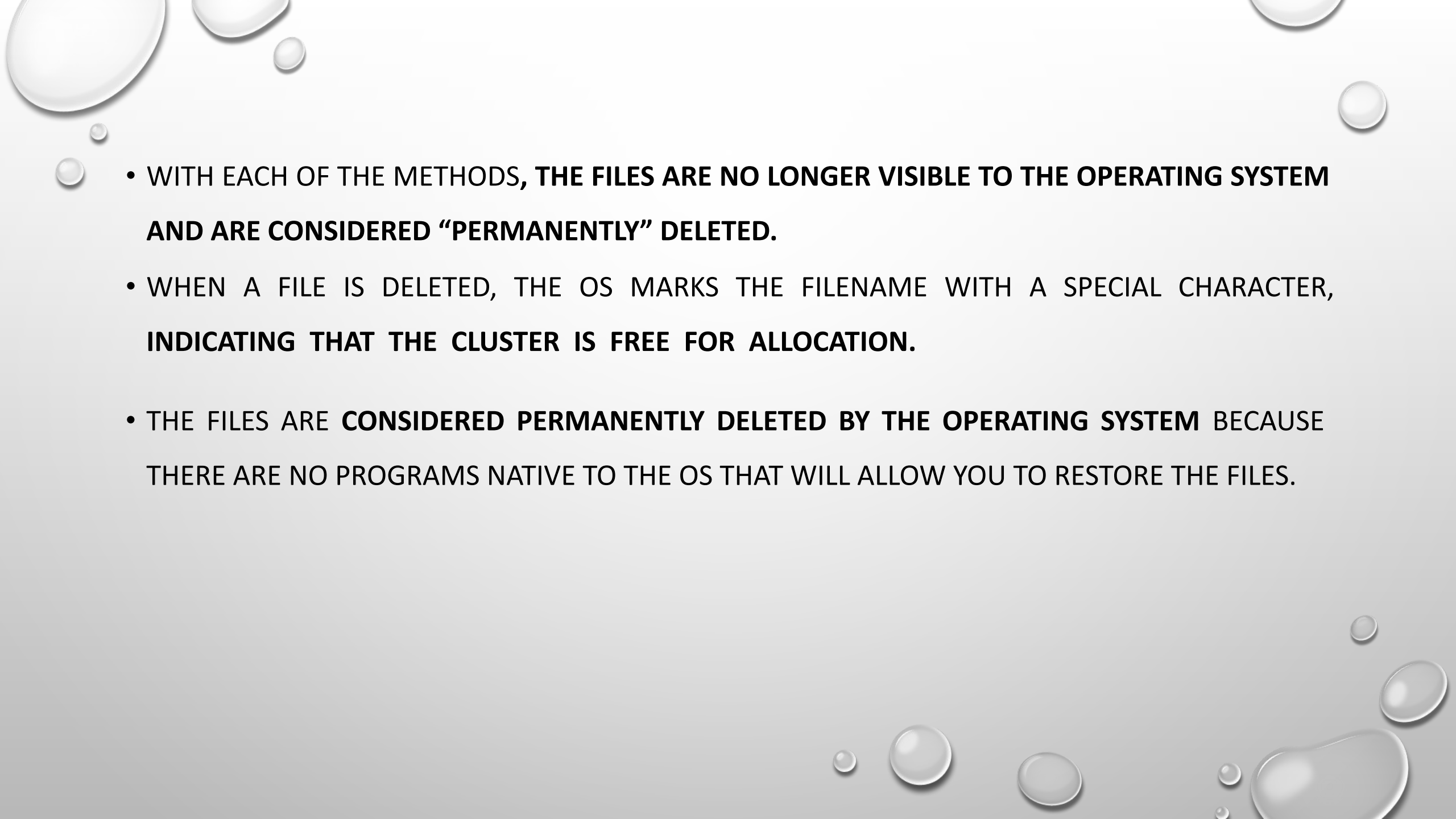
- COMMAND LINE (DEL OR ERASE)

- MOVING A FILE

- DISK CLEANUP

- WIPING/OVERWRITING



- 
- WITH EACH OF THE METHODS, THE FILES ARE NO LONGER VISIBLE TO THE OPERATING SYSTEM AND ARE CONSIDERED “PERMANENTLY” DELETED.
 - WHEN A FILE IS DELETED, THE OS MARKS THE FILENAME WITH A SPECIAL CHARACTER, INDICATING THAT THE CLUSTER IS FREE FOR ALLOCATION.
 - THE FILES ARE **CONSIDERED PERMANENTLY DELETED BY THE OPERATING SYSTEM** BECAUSE THERE ARE NO PROGRAMS NATIVE TO THE OS THAT WILL ALLOW YOU TO RESTORE THE FILES.

- IN FAT FILE SYSTEM, THE FIRST LETTER OF THE FILENAME IS REPLACED WITH HEX BYTE **E5H**.
- IN NTFS FILE SYSTEM, THE OS **MARKS THE INDEX FIELD OF MFT WITH A SPECIAL CODE WHEN FILE IS DELETED**. THIS ALLOCATES THE CLUSTER AS FREE.
- **THE CLUSTERS REMAIN FREE FOR ALLOCATION AS LONG AS NO OTHER FILE IS OVERWRITTEN ON THEM.**
- THE ACTUAL FILE REMAINS IN THE DISK.
- **THE DELETED FILE MAY GET STORED IN THE RECYCLE BIN FOR A TEMPORARY TIME PERIOD.**

RECYCLE BIN

- **A *RECYCLE BIN* OR *TRASH CAN* IS A REPOSITORY WHERE FILES ARE TEMPORARILY STORED AFTER THEY ARE DELETED.**
- WHEN A FILE IS DELETED FROM THE DESKTOP OR USING WINDOWS EXPLORER, THE FILE IS MOVED TO THE RECYCLE BIN WITH A RECORD OF ITS ORIGINAL LOCATION.
- **THE FILE HASN'T ACTUALLY BEEN DELETED; ITS POINTER HAS SIMPLY BEEN CHANGED TO SHOW THAT IT NOW RESIDES IN THE RECYCLE BIN.**
- DELETING OR EMPTYING FILES FROM THE RECYCLE BIN PERMANENTLY DELETES THEM FROM THE SYSTEM, SO THEY CAN BE RECOVERED ONLY WITH DATA RECOVERY OR FORENSIC SOFTWARE.

DEPENDENDING ON THE VERSION OF WINDOWS, RECYCLE BIN EVIDENCE IS STORED IN TWO DIFFERENT WAYS:

FOR WINDOWS XP, THE FILES ARE STORED IN THE “**RECYCLER**” FOLDER UNDER THE USER’S SPECIFIC SID.

- THERE IS ALSO AN **INFO2 FILE** WHICH CONTAINS AN **INDEX OF ALL THE FILES THAT HAVE BEEN DELETED, ALONG WITH SOME METADATA ABOUT THE RECYCLED FILES.**
- **THE INFO2 FILE WILL CONTAIN THE ORIGINAL PATH, FILE SIZE, AND WHEN THE FILE WAS DELETED.**

FOR **WINDOWS VISTA+**, THE DATA IS STILL STORED UNDER THE USER’S SID, BUT THE PARENT FOLDER IS NOW CALLED “**\$RECYCLE.BIN**”.

- WINDOWS NO LONGER USES THE INFO2 FILE AND WHEN A FILE IS DELETED, **TWO FILES ARE CREATED IN THE RECYCLE BIN.**
- THE FIRST FILE BEGINS WITH THE VALUE “\$R” FOLLOWED BY A RANDOM STRING – THIS FILE CONTAINS THE **ACTUAL CONTENTS OF THE RECYCLED FILE**
- THE SECOND FILE BEGINS WITH “\$I” AND ENDS IN THE SAME STRING AS THE “\$R” FILE – THIS FILE CONTAINS THE **METADATA FOR THAT SPECIFIC FILE .**
- THE “\$I” FILE CONTAINS THE **ORIGINAL FILENAME, PATH, FILE SIZE, AND WHEN THE FILE WAS DELETED.**

CONCEPT OF DELETED DATA RECOVERY

- WHEN THE FILE IS DELETED, THE **OPERATING SYSTEM REMOVES THE POINTERS TO THE FILE TO LOCATE THAT DATA PHYSICALLY ON THE DISK, SO THAT THE SECTOR OF THE HARD DISK, WHICH CONSISTED OF THE FILE IS NOW AVAILABLE FOR STORING OF NEW DATA.**
- HOWEVER, THE DELETED FILE STILL EXISTS IN THE STORAGE DEVICE OR HARD DISK OF THE SYSTEM, FROM WHERE IT IS POSSIBLE TO RECOVER, UNTIL BEING OVERWRITTEN BY ANOTHER FILE.
- **OVERWRITING MODIFIES THE EXISTING BINARY COMBINATION OF THE OLD FILE.**
- IF THE SECTORS ARE PARTIALLY OVERWRITTEN WITH NEW DATA, THEN THE DELETED DATA CAN BE RECOVERED PARTIALLY AND NOT FULLY.

- FILES ARE NOT IMMEDIATELY ERASED FROM THE HARD DRIVE BECAUSE IT TAKES MORE TIME TO ERASE THE DATA THAN REMOVING THE POINTERS, WHICH SLOWS DOWN THE PERFORMANCE OF THE SYSTEM.
- TO RESTORE THE FILE, YOU NEED TO USE TOOLS THAT WILL SEARCH THE HARD DISK FOR DELETED FILES AND ALLOW YOU TO UNDELETE THEM.
- **TOOLS FOR DELETED DATA/PARTITION RECOVERY:** RECUVA, PURAN FILE RECOVERY, EASEUS DATA RECOVERY.



Select the files you want to Recover by ticking the boxes and then pressing Recover.
For the best results, restore the files to a different drive.

[Switch to advanced mode](#)

☐	Filename	Path	Last Modified	Size	State	Comment
☐	[000047].docx	H:\?	Unknown	30 KB	Excellent	No overwritten clusters detected.
☐	[000048].docx	H:\?	Unknown	30 KB	Excellent	No overwritten clusters detected.
☐	[000049].docx	H:\?	Unknown	30 KB	Excellent	No overwritten clusters detected.
☐	[000050].docx	H:\?	Unknown	30 KB	Excellent	No overwritten clusters detected.
☐	[000051].docx	H:\?	Unknown	34 KB	Excellent	No overwritten clusters detected.
☐	[000052].docx	H:\?	Unknown	34 KB	Excellent	No overwritten clusters detected.
☐	[000053].docx	H:\?	Unknown	34 KB	Excellent	No overwritten clusters detected.
☐	[000054].docx	H:\?	Unknown	34 KB	Excellent	No overwritten clusters detected.
☐	[000055].docx	H:\?	Unknown	34 KB	Excellent	No overwritten clusters detected.
☐	[000056].pdf	H:\?	Unknown	158 KB	Excellent	No overwritten clusters detected.
☐	[000057].docx	H:\?	Unknown	3,389 KB	Excellent	No overwritten clusters detected.
☐	[000058].docx	H:\?	Unknown	3,752 KB	Excellent	No overwritten clusters detected.
☐	[000059].docx	H:\?	Unknown	3,908 KB	Excellent	No overwritten clusters detected.

Recovery completed in 0.25 second. Total recovered 1 file(s): fully recovered 1 file(s), partly recovered 0 file(s).

[Recover...](#)

DELETED PARTITIONS

- A COMMON TECHNIQUE THAT IS USED TO THWART A FORENSIC INVESTIGATION IS **TO REPARTITION A DISK OR CLEAR THE PARTITION STRUCTURES SO THAT THE ORIGINAL STRUCTURE IS GONE.**

PARTITIONS IN THE HARD DISK CAN GET DELETED OR LOST DUE TO ONE OF THE FOLLOWING REASONS:

- **ACCIDENTAL DELETION**
- **DATA CORRUPTION**
- **SUDDEN POWER OUTAGE**
- **INTENTIONAL USE OF TOOLS SUCH AS FDISK, DISKPART OR THE DISK MANAGEMENT FACILITY**

MAGIC NUMBERS

- ALSO KNOWN AS **FILE SIGNATURES**.
- MAGIC NUMBER IS A CONSTANT **USED TO IDENTIFY A FILE FORMAT**.
- DETECTING SUCH CONSTANTS IN FILES IS A SIMPLE WAY OF DISTINGUISHING BETWEEN FILE FORMATS, **BASICALLY EVERY FILE HAS AN HEADER AND A FOOTER IN ORDER TO GET CORRECTLY RECOGNIZED**.
- **FOR EXAMPLE** :A PDF FILE STARTS WITH “%PDF” AND ENDS WITH “%EOF” WHILE A JPEG IMAGE FILE BEGINS WITH “0xFFD8” AND ENDS WITH “0xFFD9”.

DATA CARVING

- **IT IS THE TECHNIQUE OF REASSEMBLING FILES FROM RAW DATA FRAGMENTS WHEN NO FILESYSTEM METADATA IS AVAILABLE.**

DATA CARVING MIGHT BE CLASSIFIED AS BASIC AND ADVANCED; WITH BASIC DATA CARVING IT IS ASSUMED THAT:

- **THE BEGINNING OF FILE IS NOT OVERWRITTEN**
- **THE FILE IS NOT FRAGMENTED**
- **THE FILE IS NOT COMPRESSED**

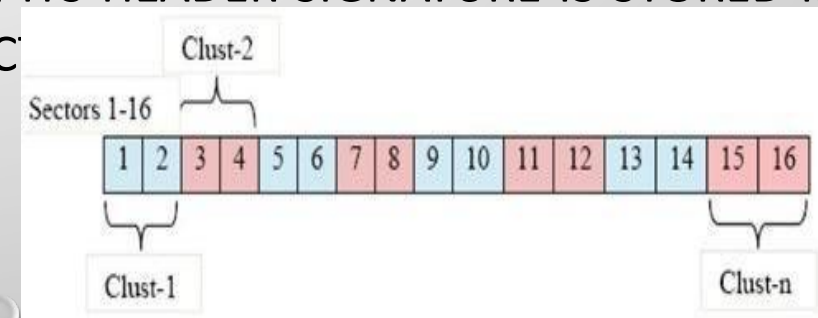
ADVANCED DATA CARVING OCCURS EVEN TO FRAGMENTED FILES, WHERE FRAGMENTS ARE:

- **NOT SEQUENTIAL**
- **OUT OF ORDER**
- **MISSING**

- THE PROCESS OF RETRIEVING INACCESSIBLE, FORMATTED, DELETED, DAMAGED OR CORRUPTED DATA FROM A STORAGE MEDIUM WHEN IT IS NOT ACCESSIBLE THROUGH NORMAL METHODS IS CALLED **DATA RECOVERY**.
- **TRADITIONAL DATA RECOVERY METHODS RELY ON FILE SYSTEM STRUCTURES LIKE FILE TABLES TO RECOVER DATA THAT HAS BEEN DELETED.**
- HOWEVER, IF THE FILESYSTEM METADATA ARE NOT AVAILABLE, CORRUPT, OR HAVE BEEN DELIBERATELY REMOVED, THE DATA WHILE PRESENT, CANNOT BE ACCESSED VIA TRADITIONAL MEANS.
- **FILE/DATA CARVING IS A FORENSICS TECHNIQUE THAT RECOVERS FILES FROM RAW DATA PRESENT ON THE MEDIA,** RATHER THAN A FILESYSTEM'S STRUCTURE OR OTHER META-DATA WHICH IS USED TO ORGANIZE DATA ON STORAGE MEDIA.

DATA CARVING TECHNIQUES

- TRADITIONAL FORENSIC DATA CARVING TOOLS AVAILABLE TODAY **SEARCH THE HEADER SIGNATURE OF A FILE** IN THE WHOLE FORENSIC IMAGE OF A GIVEN DIGITAL MEDIA.
- **ANY FILE THAT IS NEWLY CREATED WOULD BE ALLOCATED A FEW FRESH CLUSTERS OR SECTORS AND ITS HEADER SIGNATURE WILL BE AVAILABLE IN THE FIRST FEW BYTES OF FIRST CLUSTER OR SECTOR.**
- SUPPOSE A HARD DISK IS FORMATTED USING A FAT32 FILE SYSTEM AND ITS CLUSTER SIZE IS 2 SECTORS. IF A FILE IS STORED IN THE SECOND CLUSTER, ITS HEADER SIGNATURE IS STORED IN FIRST FEW BYTES OF THE SECOND CLUSTER OR THIRD SECTOR.
- **TOOLS FOR CARVING: FTK, ENCASE, PHOTOREC, ETC**



HEADER/FOOTER CARVING

- THIS METHOD OF CARVING FILES IS USED **WHEN A FILE HAS DEFINED HEADER AND FOOTER.**
- **LOCATIONS WHERE HEADER AND FOOTER VALUES ARE FOUND IN THE DATA SET ARE MARKED AND THEN ALL DATA BLOCKS BETWEEN THESE TWO LOCATIONS ARE STORED IN AN OUTPUT FILE.**
- THIS TECHNIQUE ASSUMES THAT:
 - i) **HEADER AND FOOTER VALUES OF A FILE ARE NOT DAMAGED;**
 - ii) **THE FILE IS STORED IN CONTIGUOUS DATA BLOCKS**

HEADER/EMBEDDED LENGTH CARVING

- THIS METHOD OF CARVING FILES IS USED WHEN A FILE HAS A DISTINCT HEADER AND ITS LENGTH (FILE SIZE) IS STORED IN ITS FIRST FEW BYTES.
- FILES THAT FALL IN THIS CATEGORY ARE **BMP, PDF, DAT, AVI, ETC.**
- THIS IS ONE OF THE FASTEST CARVING METHODS USED BY A CARVING TOOL.

EXAMPLE: CARVING PDF FILES

- **STEP1.** LOOK FOR THE HEADER SIGNATURE (%PDF)
- **STEP2.** CHECK FOR THE VERSION NUMBER [FILE OFFSET 6-8]
- **STEP3.** IF VERSION NO. > 1.1 GO TO STEP4. ELSE GO TO STEP6

- **STEP4.** SEARCH FOR THE STRING “LINEARIZED” IN FIRST FEW BYTES OF THE FILE.
- **STEP5.** IF IT FINDS THE STRING IN STEP 4, THEN LENGTH OF THE FILE IS PRECEDED BY A “/L ” CHARACTER SEQUENCE AS SHOWN IN FIGURE.
- **STEP6.** USE SEARCH ALGORITHMS TO FIND FOOTER SIGNATURE (%%EOF).

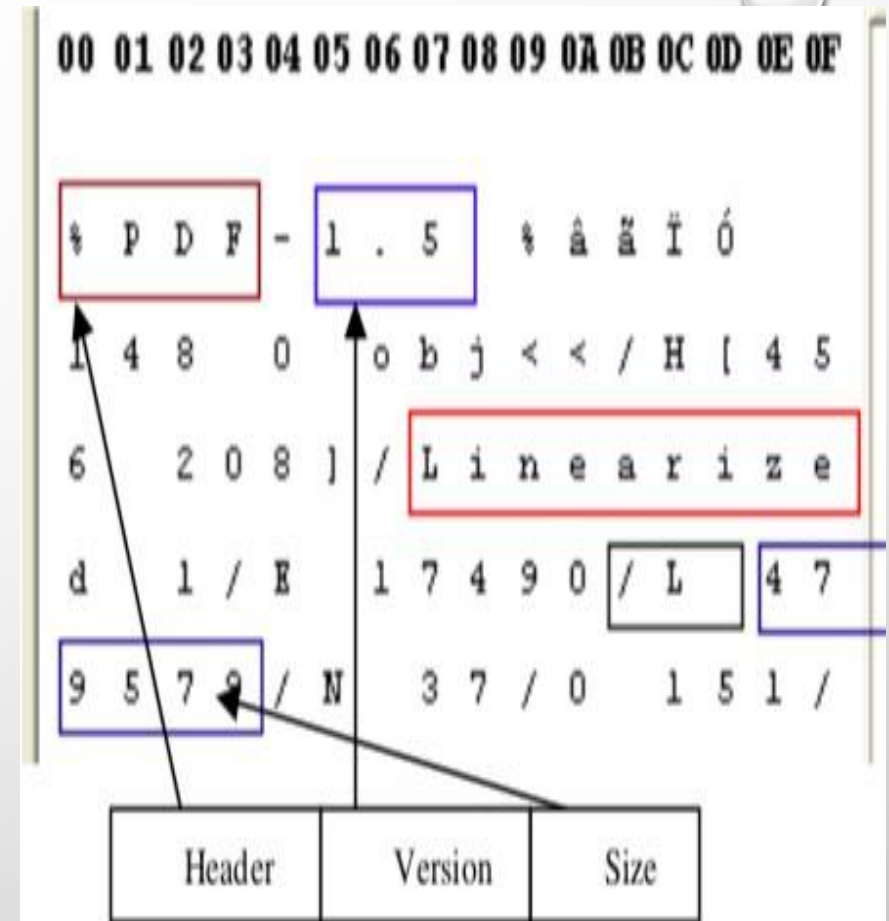


Fig. 4. Linearized pdf file header

HEADER/MAXIMUM FILE SIZE CARVING

THIS METHOD OF CARVING FILES IS USED WHEN A **FILE HAS A DISTINCT HEADER AND DOES NOT HAVE FOOTER SIGNATURE (OR FILE THAT HAS FOOTER, GETS CORRUPTED OR OVERWRITTEN).**

- IT WORKS IN SIMILAR MANNER TO THE HEADER – FOOTER CARVING.
- **FIRST THE START OF THE FILE IS LOCATED BY SEARCHING ITS HEADER. THEN END OF THE FILE IS CALCULATED BASED ON ITS SIZE.**
- **FOR SOME FILE FORMATS, THE SIZE VALUE IS SPECIFIED IN THE HEADER OF THE FILE OR THE CARVING FILE SIZE IS LIMITED BY THE USER SPECIFIED FILE SIZE.**
- ONCE THE START AND END LOCATIONS OF A FILE ARE DETERMINED, DATA BLOCKS BETWEEN THESE LOCATIONS ARE STORED IN AN OUTPUT FILE.

FILE STRUCTURE BASED CARVING

- ALSO SOMETIMES REFERRED TO AS **“SEMANTIC CARVING”** OR **“DEEP CARVING”**.
- FILE STRUCTURE BASED CARVING WORKS BY FIRST **IDENTIFYING CERTAIN LEVEL OF INFORMATION OF A FILE FORMAT.**
- **IT USES THE INTERNAL LAYOUT OF A FILE TO DETERMINE WHICH DATA IS PART OF WHICH FILE.**
- THIS LAYOUT INFORMATION IS DERIVED FROM FILE FORMAT SPECIFICATIONS, WHICH DESCRIBE THE OFFICIAL MAKEUP OF A FILE FORMAT.
- **THIS INFORMATION IS THEN MATCHED IN THE RAW DATA SET TO IDENTIFY A FILE.**
- FILE STRUCTURE BASED CARVING REDUCES FALSE POSITIVES GREATLY AS MORE INFORMATION OF THE FILE FORMAT IS USED.

BLOCK CONTENT BASED CARVING

- BLOCK BASED CARVING IS BASED ON THE PRINCIPLE THAT PHYSICAL MEDIA LIKE HARD DISKS WRITE THEIR DATA IN SECTORS (BLOCKS OF 512 BYTES), WHICH MEANS THAT FRAGMENTATION WILL ONLY OCCUR ON THESE SECTOR BOUNDARIES.
- **EACH INDIVIDUAL BLOCK IN THE DATA SET IS READ AND ITS CONTENTS ARE ANALYSED TO FIND OUT IF IT BELONGS TO A PARTICULAR FILE.**
- THE NEXT STEP IS TO PARSE EACH BLOCK AND CALCULATE META INFORMATION LIKE CHARACTER COUNTS OR STATISTICAL INFORMATION OVER THE BYTES IN A BLOCK.
- **THESE INDIVIDUAL BLOCKS ARE THEN REARRANGED IF NECESSARY TO CONSTRUCT THE ORIGINAL FILE.**

FORENSIC IMPORTANCE OF CARVING

- FILE CARVING IS A PARTICULARLY POWERFUL TECHNIQUE BECAUSE COMPUTER FILES CAN BE RECOVERED FROM RAW DATA REGARDLESS OF THE TYPE OF FILE SYSTEM, AND FILE RETRIEVAL IS POSSIBLE EVEN IF THE FILE SYSTEM METADATA HAS BEEN COMPLETELY DESTROYED.
- THE PROCESS THEREFORE PROVIDES ADDITIONAL DATA RECOVERY METHODS TO AUGMENT DIGITAL INVESTIGATION WHERE EXISTING **TRADITIONAL DATA RECOVERY TECHNIQUES ARE NOT SUITABLE OR HAVE BEEN UNSUCCESSFUL.**